COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures And Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	09/04/26	**DATE SUBMITTED:**	09/0426
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HOA 7.1 Sorting Algorithms

1. PROCEDURE OUTPUT

Table 7-1. Array of Values for Sort Algorithm Testing

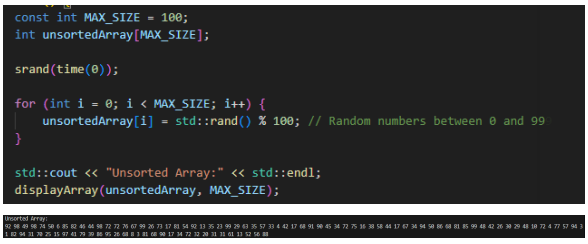
Code + Console Screenshot	
Observations	There's nothing to explain here its just a generation of unsorted Array.

Table 7-2. Bubble Sort Technique

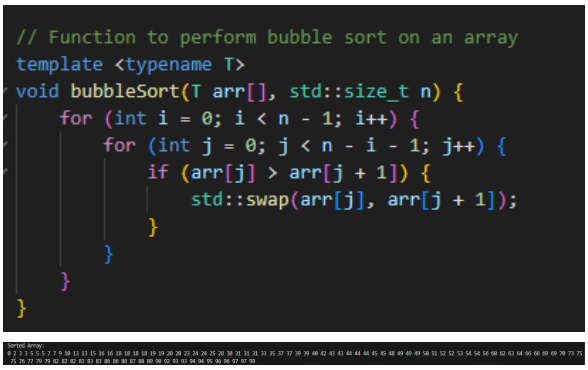
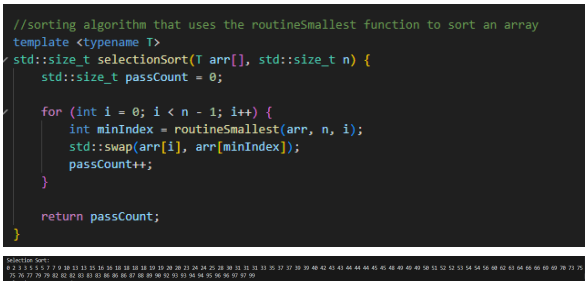
Code + Console Screenshot	
Observations	If the item on the left is larger than the item on the right, the algorithm swaps them. It does this from the beginning to the end of the array over and over again

Table 7-3. Selection Sort Algorithm

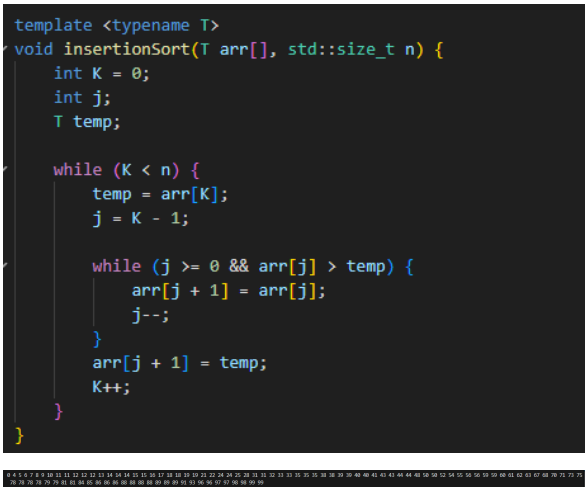
Code + Console Screenshot	
Observations	It find the smallest item once it finds it, it pulls that item out and places it at the very front of the array. It then repeats this exact process for the remaining items

```
//sorting algorithm that uses the routineSmallest function to sort an array
template <typename T>
std::size_t selectionSort(T arr[], std::size_t n) {
    std::size_t passCount = 0;

    for (int i = 0; i < n - 1; i++) {
        int minIndex = routineSmallest(arr, n, i);
        std::swap(arr[i], arr[minIndex]);
        passCount++;
    }

    return passCount;
}
```

Table 7-4. Insertion Sort Algorithm

Code + Console Screenshot	
Observations	It start first item, and as you look at next new item, you compare it to the ones you have already sorted

```
template <typename T>
void insertionSort(T arr[], std::size_t n) {
    int K = 0;
    int j;
    T temp;

    while (K < n) {
        temp = arr[K];
        j = K - 1;

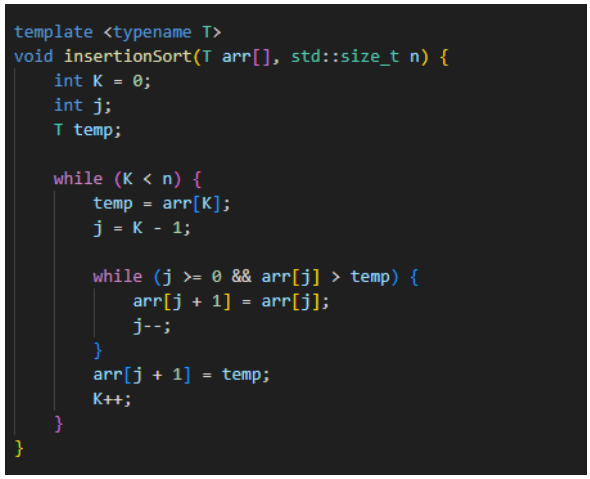
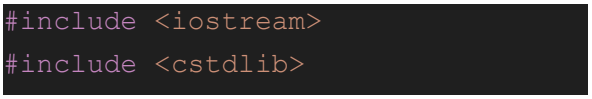
        while (j >= 0 && arr[j] > temp) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = temp;
        K++;
    }
}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Problem: Generate an array A[0…100] of unsorted elements, wherein the values in the array are indicative of a vote to a candidate. This means that the values in your array must only range from 1 to 5. Using sorting and searching techniques, develop an algorithm that will count the votes and indicate the winning candidate.

Pseudocode of Algorithm	Initialize an array votes of size 100. Loop 100 times: Assign a random number between 1 and 5 to the

	current array index. Call insertionSort on the votes array. Initialize tracking variables: currentCandidate = votes[0], currentCount = 1 winningCandidate = votes[0], maxVotes = 1 Loop from index 1 to 99 in the votes array: If votes[index] equals currentCandidate, increment currentCount. Else (the candidate number changed): If currentCount > maxVotes, update maxVotes and winningCandidate. Reset currentCandidate = votes[index] and currentCount = 1. Check one final time if the last candidate tallied has currentCount > maxVotes. Print the winning candidate's name based on the winningCandidate number.

Screenshot of Algorithm Code	
	template <typename T>
	void insertionSort(T arr[], std::size_t n) {
	int K = 0;
	int j;
	T temp;
	while (K < n) {
	temp = arr[K];
	j = K - 1;
	while (j >= 0 && arr[j] > temp) {
	arr[j + 1] = arr[j];
	j--;
	}
	arr[j + 1] = temp;
	K++;
	}
	}
	 
	#include <iostream>
	#include <cstdlib>

```cpp
#include <ctime>
#include "SortingAlgo.h"

// case function to get candidate name
std::string name(int choice) {
    switch(choice) {
        case 1: return "Bo Dalton Capistrano";
        case 2: return "Cornelius Raymon Agustin";
        case 3: return "Deja Jayla Banaga";
        case 4: return "Lalla Brielle Yabut";
        case 5: return "Franklin Relano Castro";
        default: return "Unknown";
    }
}

int main() {
    const int MAX_SIZE = 100;
    int votes[MAX_SIZE];
    srand(time(0));

    for (int i = 0; i < MAX_SIZE; i++) {
        votes[i] = (std::rand() % 5) + 1; // Random votes between 1 and 5
    }

    insertionSort(votes, MAX_SIZE);

    int currentName = votes[0];
```

```
    int currentCount = 1;

    int winningName = currentName;
    int maxVotes = 1;

    for (int i = 1; i < MAX_SIZE;
i++) {
        if (votes[i] ==
currentName) {
            currentCount++;
        } else {
            // check if previous
candidate has the most votes so
far
            if (currentCount >
maxVotes) {
                maxVotes =
currentCount;
                winningName =
currentName;
            }
            // reset for the new
candidate
            currentName =
votes[i];
            currentCount = 1;
        }
    }

    // final check for the last
candidate in the array
    if (currentCount > maxVotes) {
        maxVotes = currentCount;
        winningName = currentName;
    }

    std::cout << "--- ELECTION
```

```
RESULTS ---" << std::endl;
    std::cout << "Total votes
cast: " << MAX_SIZE << std::endl;
    std::cout << "Winning
Candidate: " << winningName <<
std::endl;
    std::cout << "Winning
Candidate Name: " <<
name(winningName) << std::endl;
    std::cout << "Total Votes
Received: " << maxVotes <<
std::endl;

    return 0;
}
```

Output Testing

```
PS C:\Programming\C++\SCHOOL\CPE 010\HOA 7p1> cd "c:\Progr
--- ELECTION RESULTS ---
Total votes cast: 100
Winning Candidate: 4
Winning Candidate Name: Lalla Brielle Yabut
Total Votes Received: 28
PS C:\Programming\C++\SCHOOL\CPE 010\HOA 7p1> cd "c:\Progr
--- ELECTION RESULTS ---
Total votes cast: 100
Winning Candidate: 4
Winning Candidate Name: Lalla Brielle Yabut
Total Votes Received: 25
PS C:\Programming\C++\SCHOOL\CPE 010\HOA 7p1> cd "c:\Progr
--- ELECTION RESULTS ---
Total votes cast: 100
Winning Candidate: 3
Winning Candidate Name: Deja Jayla Banaga
Total Votes Received: 28
PS C:\Programming\C++\SCHOOL\CPE 010\HOA 7p1> cd "c:\Progr
--- ELECTION RESULTS ---
Total votes cast: 100
Winning Candidate: 5
Winning Candidate Name: Franklin Relano Castro
Total Votes Received: 25
PS C:\Programming\C++\SCHOOL\CPE 010\HOA 7p1> cd "c:\Progr
--- ELECTION RESULTS ---
Total votes cast: 100
Winning Candidate: 4
Winning Candidate Name: Lalla Brielle Yabut
Total Votes Received: 27
PS C:\Programming\C++\SCHOOL\CPE 010\HOA 7p1>
```

Question: Was your developed vote counting algorithm effective? Why or why not?

Yes, Because the algorithm i used is Insertion Sort first, all identical votes are clustered together. This allows the searching algorithm to simply read the array from left to right

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error explanator**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **It explain to me why did that error happen**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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